ELTE
FACULTY OF
INFORMATICS

PROGRAMMING

Dynamic array

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Dynamic array

Array – the "problem"

- need to allocate memory at the beginning
- idea: only if we know the n (the size)

```
int[] ttime;
...
Console.Error.Write("n = ");
int.TryParse(Console.ReadLine(), out n);
ttime = new int[n];
```

- if we have an array in the output – what we know about the size?
  - we need to allocate the maximum – and then don't use all places
  - Multiple item selection, combined patterns,…

Example

Task: Let's **define all** excellent students in a class! List the names…

Specification:

```
In: b∈Z, e∈Z

Out: cnt∈N, y∈H[1..cnt]
Pre: -
Post: (cnt,y)=
          MIS(i=b..e,
              A(i),
              f(i))
```

Specification:

```
In: n∈N, students∈Tstudent[1..n],
    Tstudent=(name:S x grade:N)
Out: cnt∈N, exs∈S[1..cnt]
Pre: ∀i∈[1..n]:(1<=students[i].grade<=5)
Post: (cnt,exs)=
          MIS(i=1..n,
              students[i].grade=5,
              students[i].grade
```

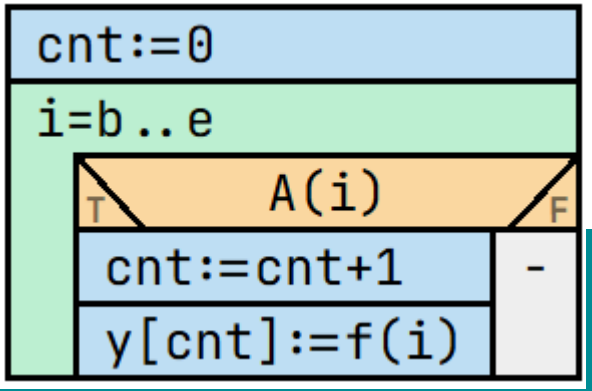
Example

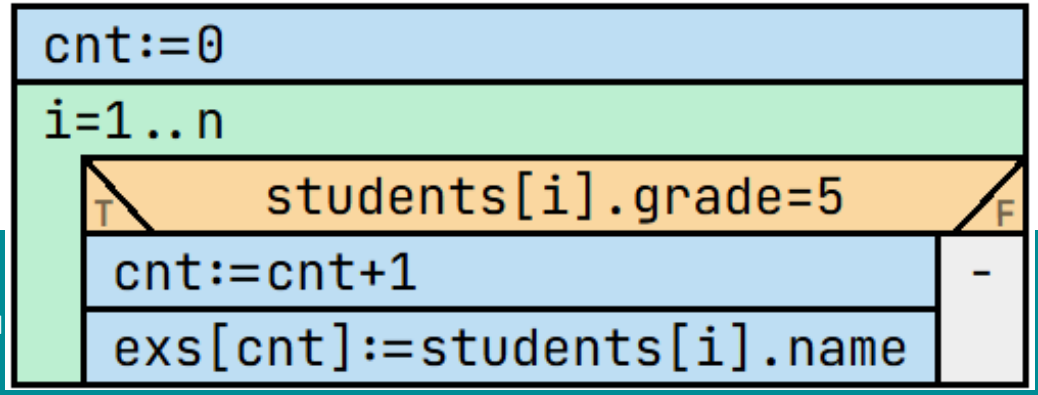
Task: Let's **define all** excellent students in a class! List the names…

cnt, y	~	cnt, exs
b..e	~	1..n
A(i)	~	students[i].grade=5
f(i)	~	students[i].name

Specification:

```
In: b∈Z, e∈Z                                  Tstudent[1..n],
                                              :S x grade:N)
Out: cnt∈N, y                                 1..cnt]
Pre: -                                        =students[i].grade<=5)
Post: (cnt,y)=                      ⟷   Post: (cnt,exs)=
          MIS(i=b..e,                                MIS(i=1..n,
              A(i),                                      students[i].grade=5,
              f(i))                                      students[i].name
```





Static array

MIS

```
struct Tstudent {
  public string name;
  public int grade;
}
static void Main(string[] args) {
  // decl.: spec. in + out
  int n; Tstudent[] students;
  int cnt; string[] exs;
  // reading the input
  Console.Write("n = ");
  int.TryParse(Console.ReadLine(), out n);
  students = new Tstudent[n];
  exs = new string[n];
  for (int i = 0; i < n; i++) {
    Console.Write("{0}. name = ", i+1);
    students[i ].name = Console.ReadLine();
    Console.Write("{0}. grade = ", i+1);
    int.TryParse(Console.ReadLine(), out students[i].grade);
  }
```

```
  // solution: stuki
  cnt = 0;
  for (int i = 0; i < n; i++) {
    if (students[i].grade == 5) {
      exs[cnt] = students[i].name;
      cnt = cnt + 1;
    }
  }
  // printing the output
  Console.WriteLine("The number of…: {0}:", cnt);
  for (int i = 0; i < cnt; i++) {
    Console.WriteLine(exs[i]);
  }
}
```

items will be only from 1 to cnt, but we allocated memory for the "worst case"

Code it!

with functions

Dynamic array

- A dynamic array is used to represent a sequence of variable number of elements
  - It can be resized during the execution as required.
  - This is exactly what we need for the multiple item selection (MIS)
- **Specification**

  Out: ~~cnt∈N,~~ y∈N[1..~~cnt~~]
  Post: ∀i∈[1..length(y)]:(A(y[i]))

- **Algorithm**

  Variable y:Array[1..~~cnt~~: Integer]

  - Code

1. get a "start address"
2. the length is 0
3. new operations:
   - length
   - at the end (Add)

```
List<int> y = new List<int>();
y.Add(1);
y.Add(2);
Console.WriteLine(y[0]);    // 1
Console.WriteLine(y.Count); // 2
```

Dynamic array – C#

- the ith item doesn't exist at the beginning

```
List<int> y = new List<int>();

y[0]=1;

y.Add(1);

y[0]=5;

y.ElementAt(0)=5;

Console.WriteLine(y.Count);
```

Doesn't exist the 1st item!

referencing to the1st item

referencing to the1st item
safeguard against a non-existent element

the length of the List

Dynamic array – the code

```
// ...
  int n; Tstudent[] students;
  List<string> exs = new List<string>(); int n;
  // reading the input
  Console.Write("n = ");
  int.TryParse(Console.ReadLine(), out n);
  students = new Tstudent[n];
    for (int i = 0; i < n; i++) {
    Console.Write("{0}. name = ", i+1);
    students[i ].name = Console.ReadLine();
    Console.Write("{0}. grade = ", i+1);
    int.TryParse(Console.ReadLine(), out students[i].grade);
  }
exs.Clear(); // int cnt = 0;
for (int i = 0; i < students.Length; i++) {
  if (students[i].grade == 5) {
    exs.Add(students[i].name);
  }
}
Console.WriteLine ("The number of…: {0}:", exs.Count);
for (int i = 0; i < exs.Count; i++) {
  Console.WriteLine(exs[i]);
}
```

Multiple item selection - dynamic

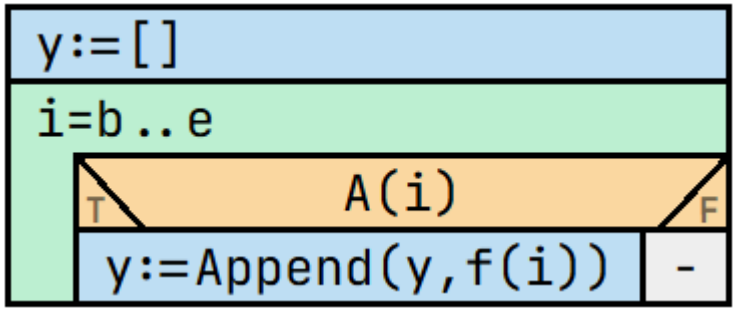
Let an interval of integers [b..e], a condition A: [b..e]→L and a function f: [b..e]→H be given.

Determine the f values of the point(s) at which A attains the "true" value on the interval [b..e]

Specification:

```
In: b∈Z, e∈Z
Out: y∈H[1..]
Pre: -
Post: (,y)=MIS(i=b..e,A(i),f(i))
```

Algorithm:



Example

Task: Let's **define all** excellent students in a class! List the names…

Specification:

```
In: b∈Z, e∈Z

Out: y∈H[1..]
Pre: -
Post: (,y)=
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Out: exs∈S[1..]
Pre: ∀i∈[1..n]:(1<=students[i].grade<=5)
Post: (,exs)=
        MIS(i=1..n,
            students[i].grade=5,
            students[i].grade
```
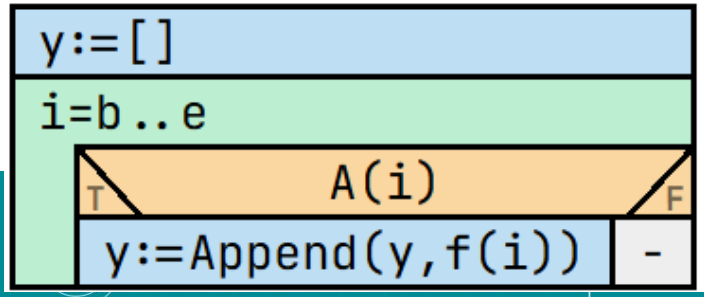
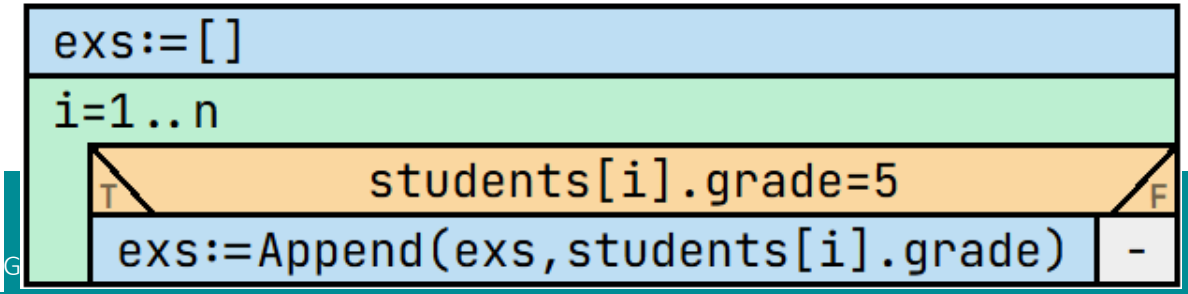

y	~	exs
b..e	~	1..n
A(i)	~	students[i].grade=5
f(i)	~	students[i].name





Dynamic array – the code

```
static void Main(string[] args) {
  // declaring (variables, specification: in/out)
  int n;
  List<int> ttime = new List<int>;

  // ...

  int n;
  Console.Error.Write("n = ");
  int.TryParse(Console.ReadLine(), out n);
  int hv;
  for (int i = 0; i < n; i++) {
    Console.Error.Write("{0}. traveling time = ", i+1);
    int.TryParse(Console.ReadLine(), out hv);
    ttime.Add(hv);
  }
}
```

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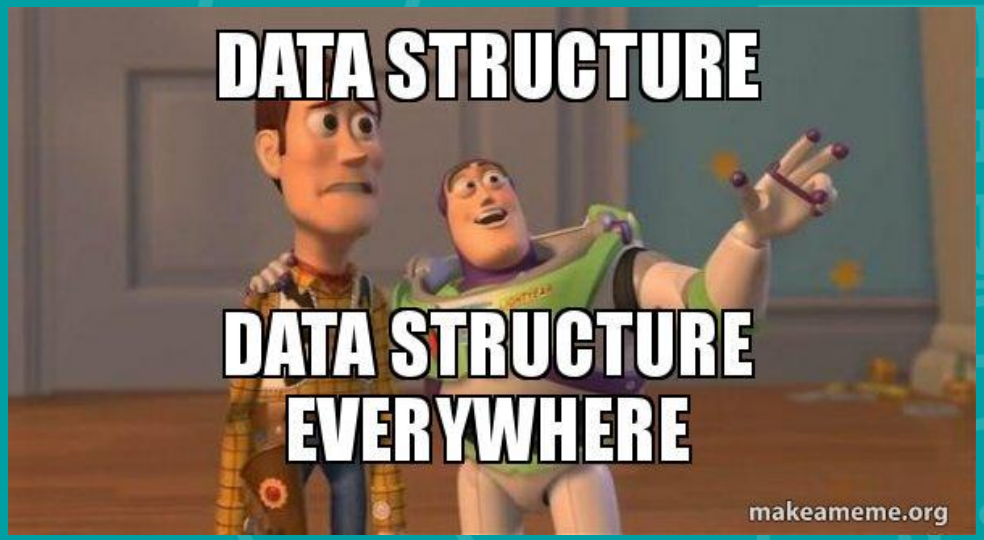

Thank you for your attention!